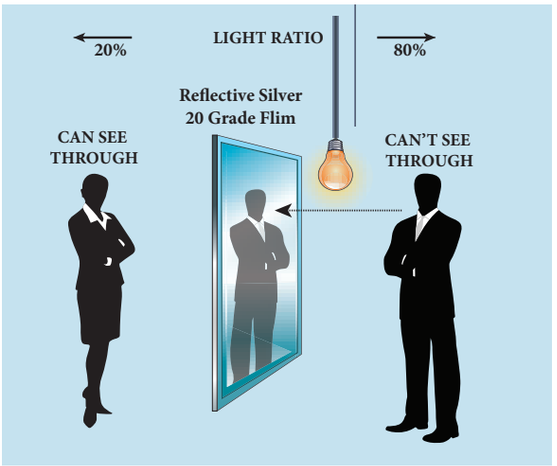
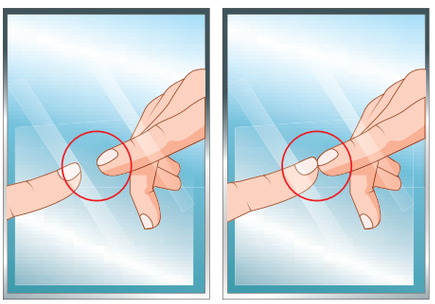


Production of optical surfaces capable of refracting as well as reflecting is possible by properly coating the surfaces with suitable materials. Thus, a glass can be made partially see through and partially reflecting. These glasses are commercially called as two-way mirror, half-silvered mirror, semi-silvered mirror etc. This gives a perception of regular mirror if the other side is made dark. But, still hidden cameras can be kept behind such mirrors. We need to be cautious when we stand in front of mirrors kept in unknown places. There is a method to test the two way mirror. Place the finger nail on the mirror surface. If there is a gap between nail and its image, then it is a regular mirror. If the fingernail directly touches its image, then it is a two way mirror.





6.4.3 Principle of reversibility

The *principle of reversibility* states that light will follow exactly the same path if its direction of travel is reversed. This is true for both reflection and refraction as shown in Figure 6.19.

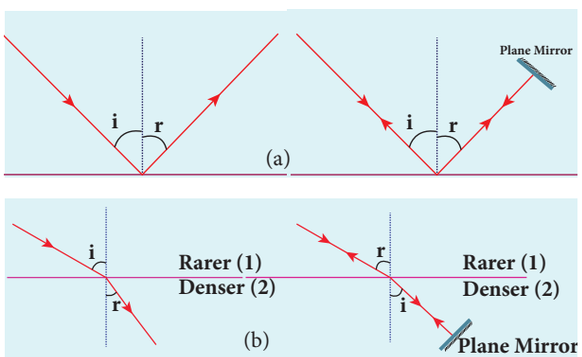


Figure 6.19 Principle of reversibility in (a) reflection and (b) refraction

6.4.4 Relative refractive index

In equation (6.18), **the term** $\left(\frac{n_2}{n_1}\right)$ **is called *relative refractive index* of second medium with respect to the first medium** which is denoted as n_{21}.

$$n_{21} = \frac{n_2}{n_1} \qquad (6.22)$$

The concept of relative refractive index gives rise to other useful relations such as,

(i) Inverse rule:

$$n_{12} = \frac{1}{n_{21}} \text{ (or) } \frac{n_1}{n_2} = \frac{1}{n_2 / n_1} \qquad (6.23)$$

(ii) Chain rule:

$$n_{32} = n_{31} \times n_{12} \text{ (or) } \frac{n_3}{n_2} = \frac{n_3}{n_1} \times \frac{n_1}{n_2} \qquad (6.24)$$

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